

Copper rod at 50 Hz

Hybrid A- Φ finite-element solver, tree-cotree gauge. A slender copper conductor driven by a 1 V terminal voltage, validated against the exact axisymmetric solution.

regression_tests/02_Ansys_Cylinder_50Hz · 240,990 unknowns ·
backward error 4.8e-21

0. How to read these figures

Every figure here comes from one solve, on one mesh — the Netgen-optimised mesh in §1. "Netgen-optimised" is a property of that mesh, not a variant of any individual plot: there is no unoptimised figure here to compare against. What Netgen changed, and by how much, is in §2.1.

Only one pair differs in *processing* rather than in what it shows: §4.2 applies one smoothing pass, and those are the only two images with a red banner burned into them.

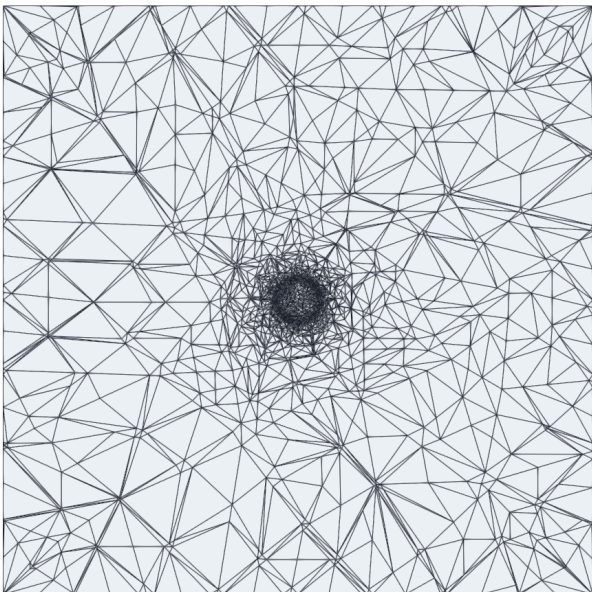
SECTION	WHAT THE FIGURE IS
§2	cross-section mesh, whole domain and the wire alone
§3	$\text{Re}(\Phi)$ and $ \Phi $ on the wire's lateral surface
§4.1	nodal B, unsmoothed — the solver's output
§4.2	the same data, one smoothing pass — not solver output
§5.1	per-node E , whole domain, log scale
§5.1.1	where the nodal field is not to be trusted (no figure)
§5.2	nodal $ E $ in the wire, auto range beside pinned range
§5.3	J as coloured vectors, longitudinal cut

1. The case

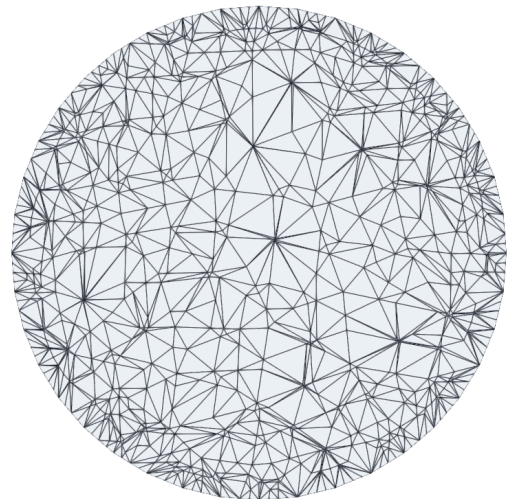
Conductor	copper, $\sigma = 5.8e7 \text{ S/m}$, $a = 1.5 \text{ mm}$, $L = 40 \text{ mm}$, a 40-gon
Domain	40 mm cube of air, $\mathbf{n} \times \mathbf{A} = \mathbf{0}$ on the outer boundary
Excitation	1.0 V top face, 0.0 V bottom face, 50 Hz
Skin depth	$\delta = 9.346 \text{ mm}$, so $a/\delta = 0.1605$ — no skin effect
Regime	$\omega L/R = 0.0725$, resistance dominated, so Φ is readable
Formulation	first-order Whitney edge $\mathbf{A}$, second-order P2 nodal Φ
Mesh	18,994 nodes, 101,873 tets, Netgen-optimised , 0.25 mm at the surface, 0.4 mm in the core
Solve	240,990 unknowns, 503 s, 4.03 GB

a/δ and $\omega L/R$ are the same parameter, both scaling as ωa^2 . A case cannot show a strong skin effect and a readable potential at once; this one takes the resistive branch.

2. Mesh



Whole domain. Graded from 0.25 mm at the conductor to 6 mm in the far air.



The wire alone. Roughly 6 elements across the radius.

The element budget sits where the field is: **6.7 %** inside $r = 1 \text{ mm}$, **84.8 %** in the band $1.0\text{--}2.0 \text{ mm}$, and **3.8 %** beyond $r = 3 \text{ mm}$.

2.1 What Netgen optimisation bought

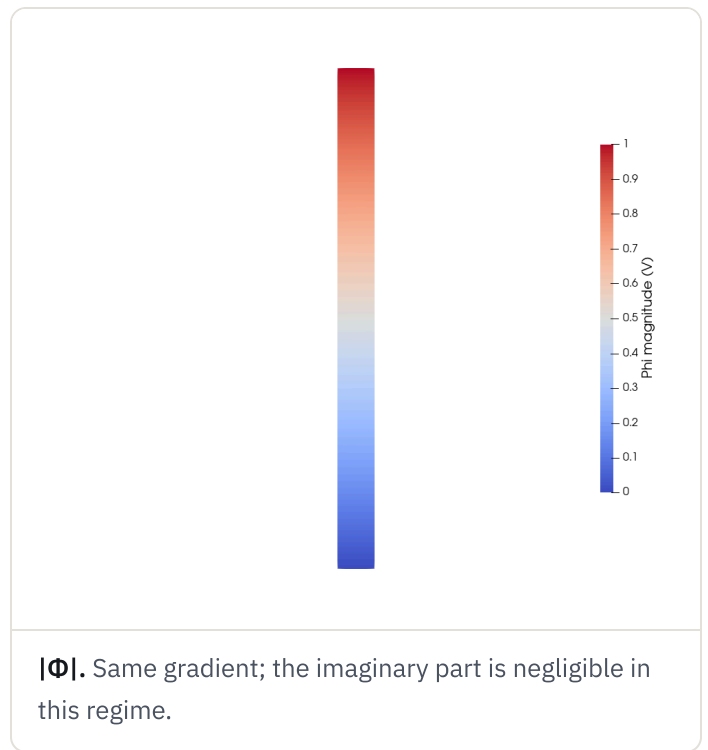
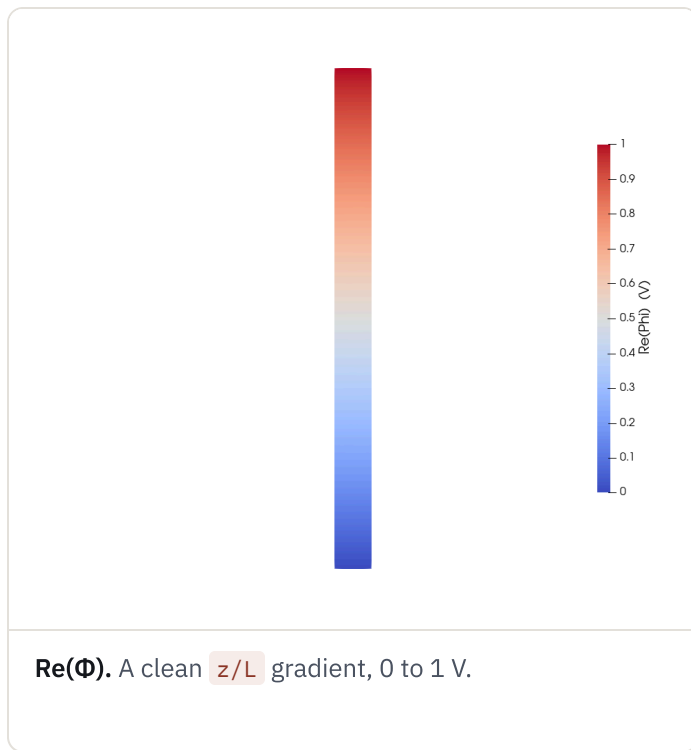
It changes element *shape* at fixed element size. Measured against the same mesh generated without it — same geometry file, same sizes, only the flag changed:

	WITHOUT	WITH NETGEN	
tets	105,715	101,873	CombineImprove merges elements
unknowns	248,294	240,990	
factorise	681 s, 4.56 GB	503 s, 4.03 GB	26 % faster, 12 % less memory
scatter at the peak	0.0181	0.0138	−24 %
scatter near the core	0.0309	0.0210	−32 %
azimuthal purity	0.9996	0.9999	

Better-shaped elements give the sparse ordering less fill-in, so the quality gain pays for itself twice. For comparison, the last *refinement* step bought 0.0215 → 0.0181 for 1.6× the unknowns and 2.8× the time. Netgen bought 0.0181 → 0.0138 for nothing.

3. Potential

Φ over the lateral surface of the wire, seen side on — not a cross-section, because the potential is driven along the axis.

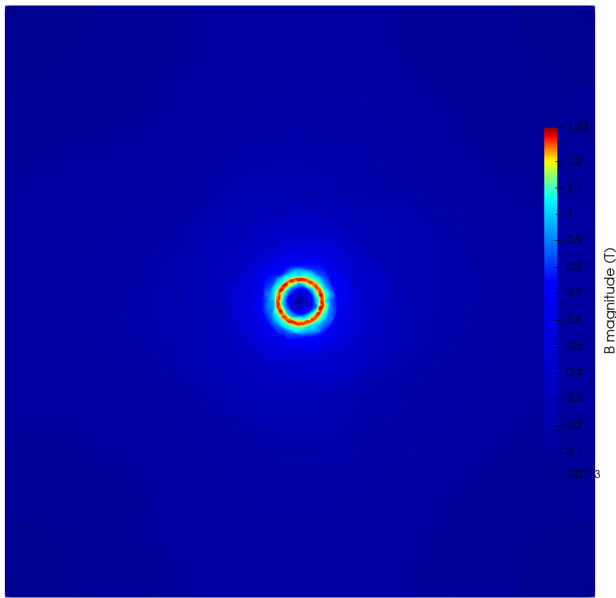


Against the exact z/L over all 95,149 wire nodes: worst deviation **8.26e-04**, mean **2.51e-04**. Φ is gauge dependent — a different spanning tree shifts it by $-j\omega\psi$, almost purely imaginary. **E**, **B**, **H** and **J** are not.

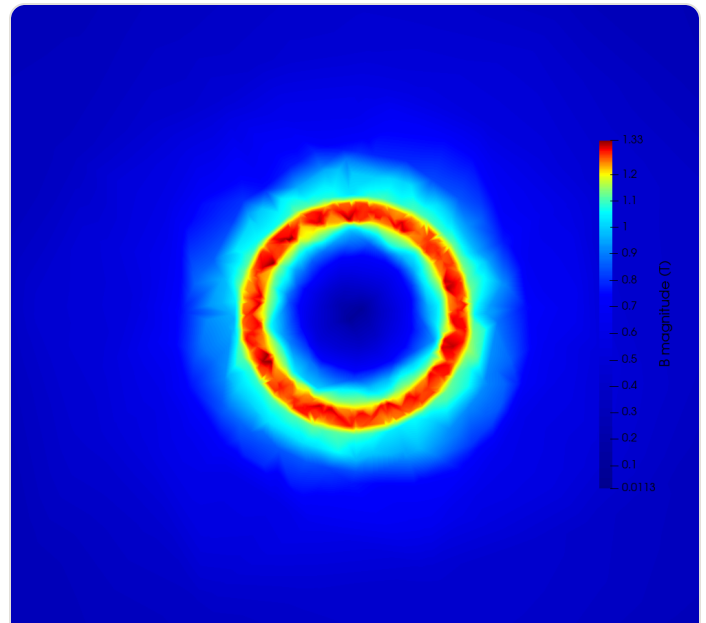
4. Magnetic flux density

4.1 The solver's output

Nodal **|B|**, mid-length cross-section, **no smoothing**. Linear colour scale — a log ramp compresses the $1/r$ decay into the top few colours and hides the ring entirely. These are the figures to read numbers off.



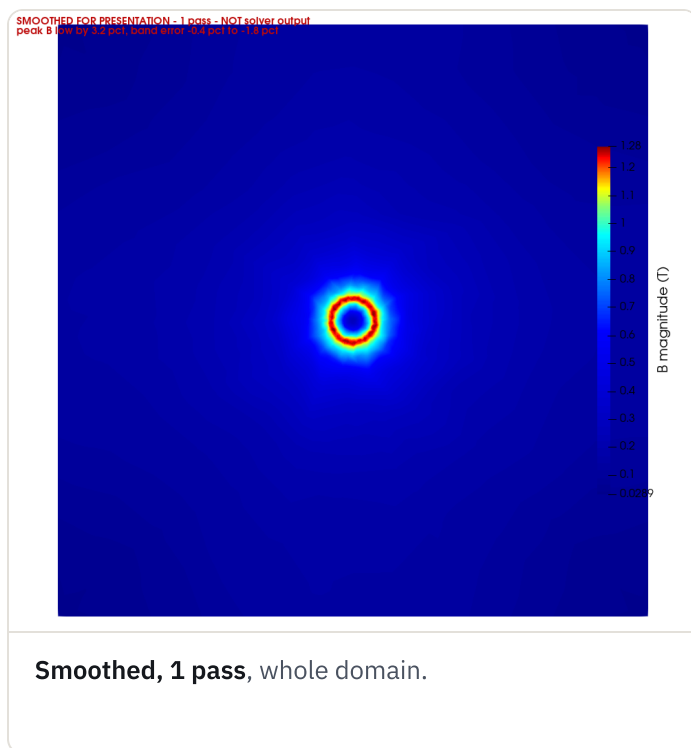
Whole domain. $1/r$ decay to the box wall. Peak 1.3385 T.



Zoomed to 3a. Zero on the axis, linear inside, peak at $r = a$.

4.2 The same data with one smoothing pass

The only difference from §4.1 is one point-to-cell-to-point averaging round trip. Same solve, same mesh, same colour map. Commercial tools smooth plotted fields by default, which is the leading explanation for why their $|B|$ cross-sections look cleaner, so these exist to compare like with like. **They are not the solver's output**, and each carries a red banner saying so.

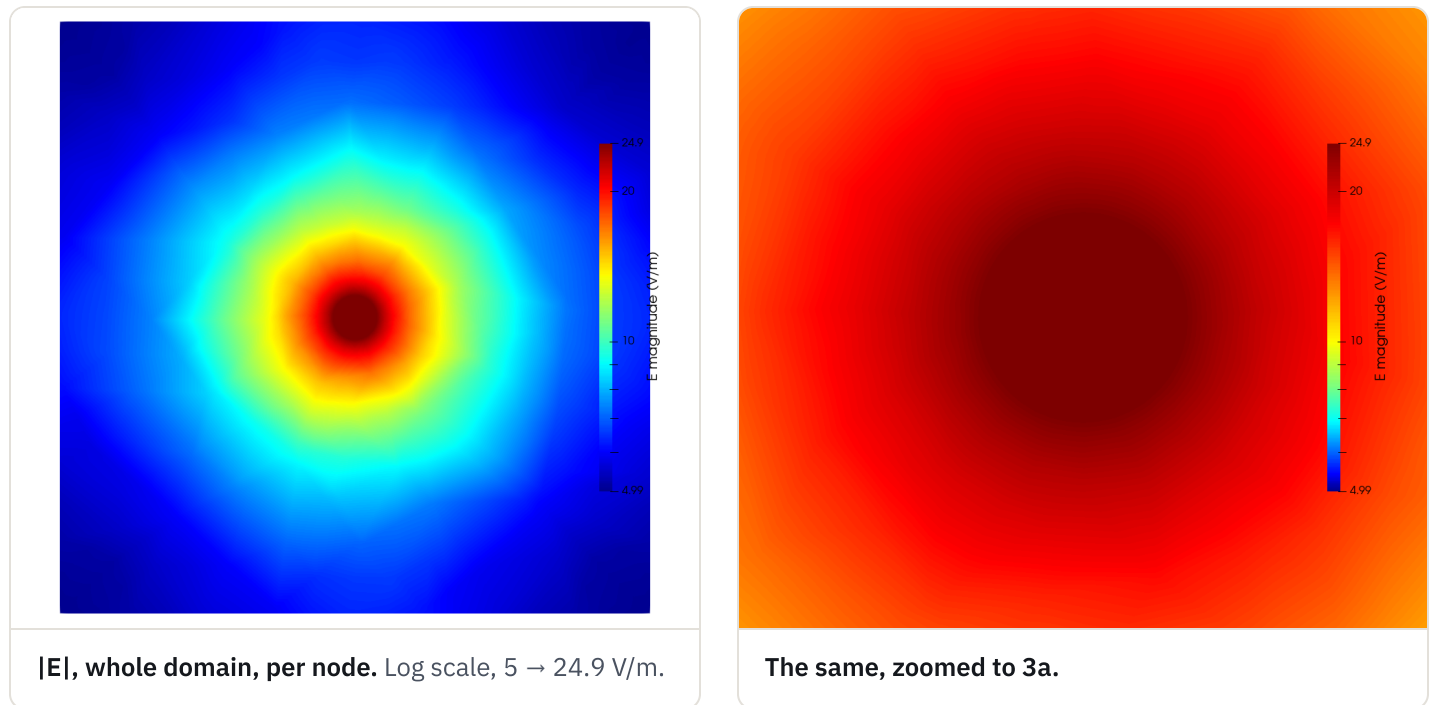


	UNSMOOTHED (§4.1)	1 PASS (§4.2)	2 PASSES	4 PASSES
scatter, 1.20–1.49 mm	0.0296	0.0183	0.0179	0.0225
error, 1.20–1.49 mm	–0.39 %	–1.84 %	–2.43 %	–3.48 %
displayed peak (T)	1.3385	1.2954	1.2716	1.2387
peak change	—	–3.2 %	–5.0 %	–7.5 %

38 % less scatter for 3.2 % of the peak — a better trade than on the pre-Netgen mesh, where it cost 6.2 % for 14 %. The bias still grows while the scatter stops improving past one pass. Smoothing is not a route to a better answer; it is the control needed to compare like with like if the reference picture is itself smoothed.

5. Electric field

5.1 Whole domain, per node



The scale is logarithmic, 5 → 24.9 V/m. The conductor is at the **top** of that range, which is correct: **E** is largest inside the copper and decays outward.

5.1.1 Where the nodal field is not to be trusted

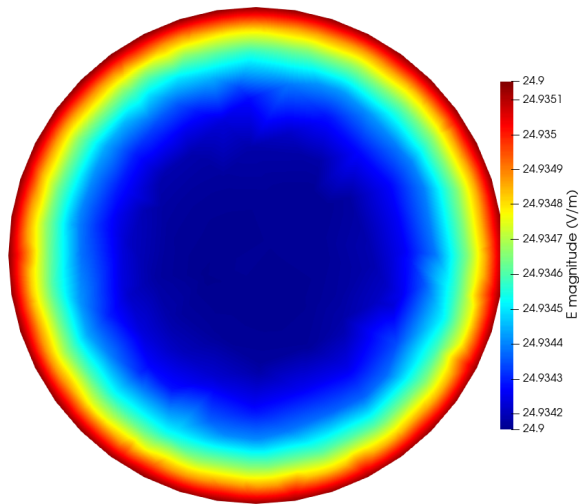
The post-processor also writes a per-cell **E**, one constant value per tetrahedron. It is not plotted here — it renders as a mosaic and reads badly — but it is the reference the nodal field is checked against, because it is exact at the material interface where the nodal field is not. Mean **|E|** by radius, nodal against per cell:

BAND (MM)	NODAL	PER CELL	DIFFERENCE
0.0 – 1.5	24.934	24.934	0.00 %
1.5 – 2.0	29.972	37.882	–20.88 %
2.0 – 3.0	29.866	29.775	+0.31 %
3.0 – 5.0	21.507	21.493	+0.06 %
5.0 – 8.0	14.807	14.690	+0.80 %

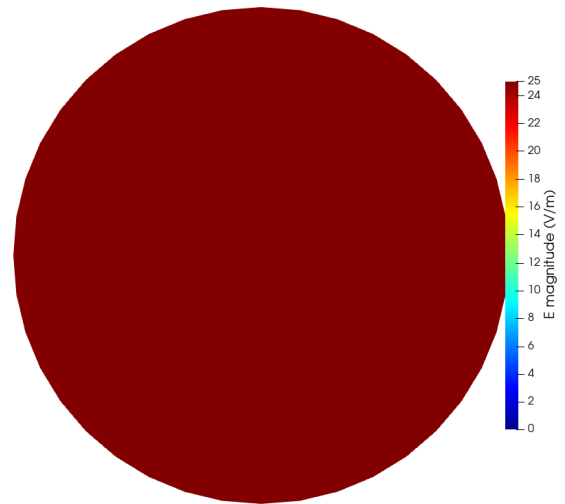
Inside the conductor the two agree exactly, and from $r = 2 \text{ mm}$ outward to better than 1 %. **Exactly one band differs — the one immediately outside the wire.** E 's *normal* component jumps at the material interface, so no single nodal value is correct there. The post-processor resolves the ambiguity by taking the higher-conductivity side, so every interface node carries the copper value, 24.93 V/m — but the air just outside genuinely carries 37.88 V/m, because a radial component appears there that the purely axial field inside does not have. $|E|$ jumps **up** crossing into the air, so forcing the copper value onto those nodes makes that first band read **21 % low**, not high.

45,200 of 140,532 nodes (32.2 %) are flagged as interface nodes, all at $r = 1.4954\text{--}1.5000 \text{ mm}$. Use the per-cell figure when the value at the surface matters; use the nodal one when the overall picture does.

5.2 Inside the wire — and why the colours disagree with §5.1



Auto colour range. Range 24.9342 → 24.9351 V/m.



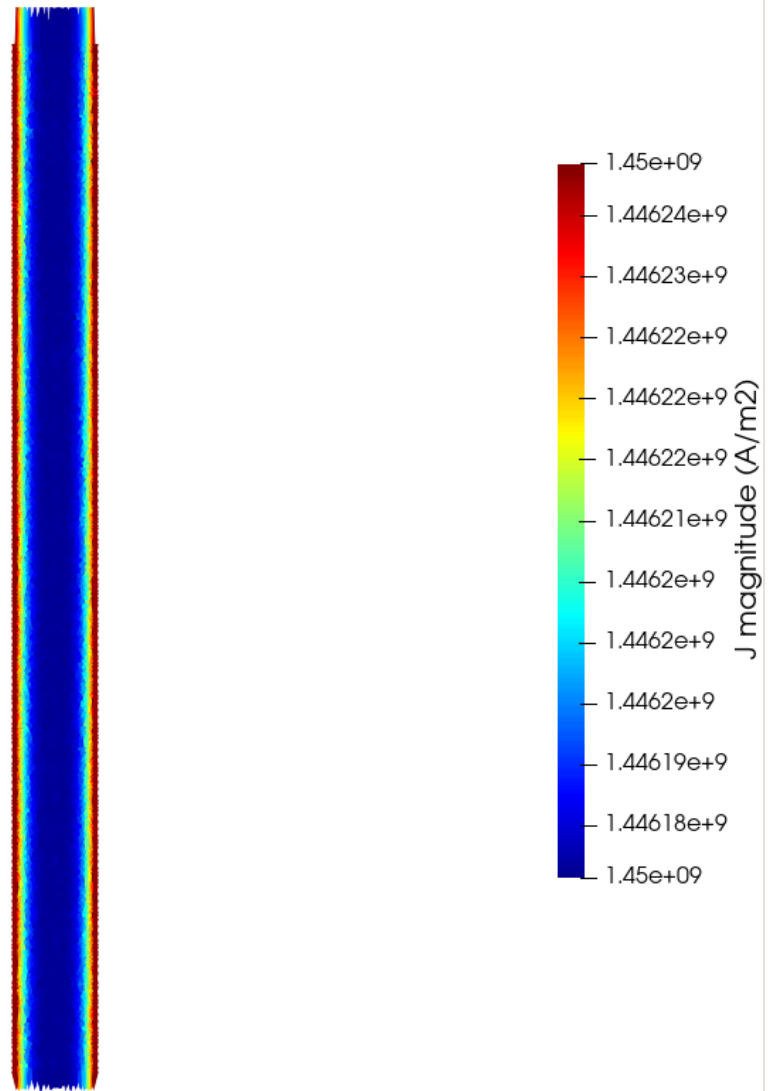
Range pinned 0–25 V/m. The same data.

This answers why the centre reads red in one figure and blue in the other. Both show the same number, about 24.935 V/m, against colour ranges that differ by a factor of roughly 20,000.

In §5.1 the range is $5 \rightarrow 24.9 \text{ V/m}$ across the whole box, so the conductor sits at the top and renders dark red. In the left figure above, ParaView rescaled to the wire's *own* range — $24.9342 \rightarrow 24.9351 \text{ V/m}$, a total span of $9\text{e-}04 \text{ V/m}$, or **0.004 %** of the value. Auto-rescaling blew that up to the full colour map, producing a red ring and blue core that look like a strong radial gradient and are in fact noise at the fifth significant figure.

The right figure pins the range to $0\text{--}25 \text{ V/m}$ and the same data renders as a uniform disc — the physical truth. With $a/\delta = 0.16$ there is no skin effect and $E = V/L$ uniformly. The exact value is 25.0 V/m; we are **0.26 %** low, consistent with the 40-gon being 0.41 % smaller in area than the circle it approximates.

5.3 Current density



$\mathbf{J} = \sigma \mathbf{E}$, on a longitudinal slice. On a mid-length cut every arrow points at the viewer and nothing is visible.

6. Validation

Nothing below fits a free parameter. The current follows from R_{dc} under the 1 V drive: $I = 10207.3 \text{ A}$, $B(a) = 1.3610 \text{ T}$.

R	9.796995e-05 Ω against 9.796863e-05 exact — 1.3e-05
L from Im(Z)/ω	22.60 nH
J(θ)/J(a)	0.999968 against the Bessel 0.999959
Φ vs exact z/L	worst 8.26e-04, mean 2.51e-04
B direction at the peak	azimuthal component 0.9999 of the total — 0.01 % off
B inside the conductor	every band within 0.5 %

BAND (MM)	MEAN, T	EXACT, T	ERROR	AZIMUTHAL SCATTER
0.30 – 0.60	0.4167	0.4163	+0.24 %	0.0477
0.60 – 0.90	0.6914	0.6884	+0.40 %	0.0309
0.90 – 1.20	0.9906	0.9929	–0.22 %	0.0210
1.20 – 1.49	1.2734	1.2722	+0.10 %	0.0138
1.49 – 2.00	1.2380	1.2633	–1.99 %	0.0307
2.00 – 3.00	0.8776	0.8869	–1.15 %	0.0559
3.00 – 5.00	0.5490	0.5601	–2.01 %	0.0768
5.00 – 8.00	0.3267	0.3349	–2.37 %	0.0868

Linear rise inside, **1/r** outside, correct absolute scale, over a 25× span in radius. Inside the conductor every band is within 0.5 %.

7. Known limitations

- **Azimuthal scatter in **B** of 1.4–3.1 % near the conductor.** The problem is axisymmetric, so this is error. It is **O(h)** element noise from **B** being constant per tetrahedron — the lowest-order quantity in the formulation.

- **Nodal B reads low in the far field — the figures in §4, not the numbers in §6.** §4 plots the nodal array; §6 is computed from the per-cell one. Against the exact solution at each sample's own radius: $0.30\text{--}1.49\text{ mm}$ nodal -0.4 to -1.5% against per-cell $+0.1$ to $+0.4\%$; then $1.49\text{--}2.00$ -4.29 vs -2.01% , $3.00\text{--}5.00$ -6.15 vs -2.12% , $5.00\text{--}8.00$ -7.50 vs -2.19% . Inside the conductor and at the peak both are within 1.5% , so the 1.3385 T peak is sound; beyond the wire the nodal colours under-read. Likely cause, reasoned rather than separately measured: the nodal value is a volume-weighted average over each node's element patch, the mesh coarsens rapidly outward, and the patch is therefore dominated by the larger outer elements where $|B| \sim 1/r$ is smaller. Unlike E , B has **no** material-interface problem here, because $\mu_r = 1$ on both sides — the band that is E 's worst is B 's best. A magnetic material would change that.
- **A coherent polygon harmonic is now visible, at 0.31% .** Fourier analysis at $r = a$ finds a peak at $m = 40$, the polygon order. No earlier mesh showed one — it was buried under element-quality noise, and Netgen removed enough of that to expose it. At 0.31% against 1.4% total scatter it is not what the eye sees, but it means raising N becomes worthwhile once quality noise falls further.
- **Far-field accuracy was traded for near-field resolution.** Beyond $r = 3\text{ mm}$ the error is $\sim 2\%$ against $\sim 1\%$ on a uniformly-graded mesh. That band holds 3.8% of the elements and does not affect R , L or the conductor fields.
- **Tried and rejected, all measured:** superconvergent patch recovery (made B more than twice as rough), plot smoothing (costs peak amplitude for modest gain), raising the polygon order alone (no effect on scatter, and slivers above $N = 48$), coarsening the far field further (a few per cent of the mesh to reclaim).
- **Uniform refinement is reaching its limit.** lc_skin $0.7 \rightarrow 0.35 \rightarrow 0.25$ gave peak-band scatter $0.029 \rightarrow 0.0215 \rightarrow 0.0181$, ratios $\times 0.74$ and $\times 0.84$ where $O(h)$ allows $\times 0.50$ and $\times 0.71$. Netgen then took it to **0.0138 for free** — a larger gain than the last refinement step bought. Element *shape* was the cheaper lever all along.